

Scalability & Future Plan

Date	28 June 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Scalability & Future Plan

This document outlines how the Credit Card Approval Prediction system can be scaled to support larger user loads, cloud-based deployment, enhanced security, and additional machine learning capabilities in future releases.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Single-server Flask deployment	Limited concurrent users	High
2	Local dataset storage	Difficult to manage large datasets	Medium
3	Basic authentication and API security	Security risks in production	High

Scalability Plan

S.No	Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users	Deploy on cloud with load balancing
2	Data Storage	Local files	Use cloud database and object storage
3	Performance	Single-thread processing	Introduce caching and optimized APIs
4	Security	Environment variables only	Add authentication, HTTPS, and monitoring

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy Flask app on cloud platform	1 Month	Public access and scalability
Phase 2	Add XGBoost hyperparameter tuning	2 Months	Improve prediction accuracy
Phase 3	Implement user authentication dashboard	3 Months	Enhanced security and tracking
Phase 4	Real-time API integration with banking systems	6 Months	Automated credit approval workflow